PHYS0525 Analog and Digital Electronics – Spring 2026

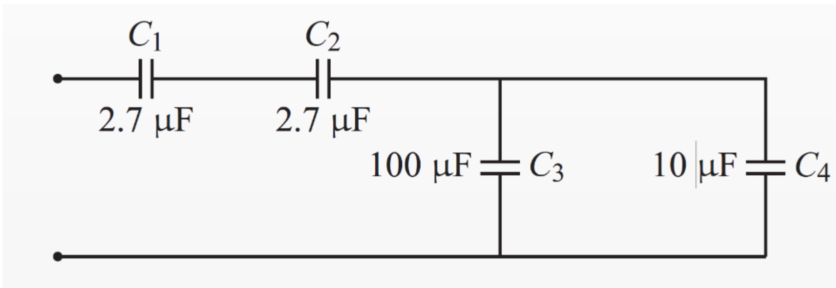
Assignment 2

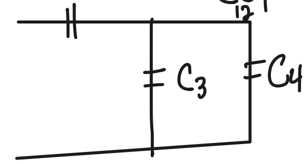
Due Monday, Feb 2, 2026

$\frac{1}{C_{12eq}} = \frac{1}{C_1} + \frac{1}{C_2} \Rightarrow \frac{1}{C_{eq}} = \frac{1}{2.7} + \frac{1}{2.7}$

1) Find the equivalent capacitance of the circuit drawn below.

$C_{eq_{12}} = \left(\frac{2}{2.7}\right)^{-1} \Rightarrow 1.35\mu F$

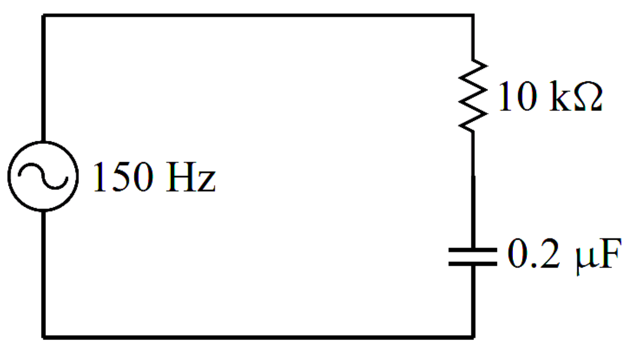




$C_{34eq} = C_3 + C_4 = 110\mu F$

$\frac{1}{C_{eq}} = \frac{1}{C_{12}} + \frac{1}{C_{34}} = \left(\frac{1}{1.35\mu F} + \frac{1}{110\mu F}\right)^{-1} = 1.33\mu F$

2) Assuming the amplitude of the voltage source in the circuit below is 1 V, what is the amplitude of the voltage across the capacitor? What is the average power dissipated by the resistor (averaged over many cycles of the voltage source)?



$V_c = V_a \cdot \frac{X_c}{Z}$

$X_c = \frac{1}{2\pi f C} = \frac{1}{2\pi(150Hz)\cdot 0.2\mu F} \Rightarrow 5305.16$

$Z = \sqrt{X_c^2 + R^2} = \sqrt{10k\Omega^2 \cdot (5305.16)^2} \Rightarrow 113210.$

$V_c = 1 \cdot \frac{5305.16}{113210.2} = 0.46816\,V$

$P_{avg} = (I_{rms})^2 \cdot R \qquad I_{rms} = I_{peak}/\sqrt{2} \qquad V_a/Z$

$= 1/113210 = 8.8331\times10^{-5} A/\sqrt{2} = (6.24\times10^{-5} A)^2 \cdot [10^4] = 3.9\times10^{-6} W = 3.9\mu W$

3) I have a circuit element with complex impedance $Z = R + \frac{1}{i\,\omega\,C}$, where $R = 100\Omega$, $C = 160\,nF$, and $\omega = 2\pi \times 10$ kHz. Compute the real and imaginary parts of Z. Also compute the magnitude and phase of Z.

$Z = \underset{\text{(Real)}}{R} + \underset{\text{(imaginary)}}{\frac{1}{i\omega C}}$

$\frac{1}{i} = -i \Rightarrow \frac{1}{i\omega C} = -i\left(\frac{1}{\omega C}\right) = -i\left(\frac{1}{2\pi\times[10\times10^3 Hz]\cdot 160\cdot[10\times10^{-9} F]}\right)$

$\Rightarrow -i(99.471)$

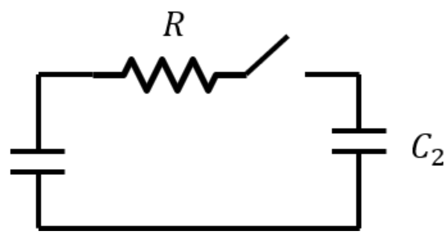
magnitude: $\sqrt{Re^2 + Im^2} \Rightarrow \sqrt{100^2 + 99.471^2} = 141.0477$

phase: $\arctan\left(\frac{Re}{-Im}\right) = \tan^{-1}\left(\frac{100\Omega}{-99.471\,AV^{-1}s^{-1}}\right) = -1.576°$

4) In the circuit below, C_1 is initially charged to a voltage of $V_{C1}(t=0) = V_0$ and C_2 initially has no charge $V_{C2}(t=0) = \ 0$ and the switch is open. At time $t = 0$ the switch is closed. Sketch $V_{C1}(t)$ and $V_{C2}(t)$ and compute the total amount of energy dissipated in the resistor. Assume the capacitance of the two capacitors is the same $C_1 = C_2 = C$.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}, \quad C_1 = C_2 : \frac{1}{C_{eq}} = \frac{2}{C}$$

$$\hookrightarrow C_{eq} = \frac{C}{2}$$



@ t_0 $V_{C_1} = V_0$

@ t_0 $V_{C_2} = 0$

$C_1 = C_2$

@ $t = \infty$, $V_{C_1}(\infty) = V_{C_2}(\infty)$

$$Q_i = Q_{C_0}, \quad Q_f = Q_{C_{1_f}} + Q_{C_{2_f}}$$

$$CV_0 = 2CV_f, \quad V_f = \frac{CV_0}{2C} = \frac{V_0}{2}$$

$$V_0 - IR - V_C = 0$$

$$v(t) = V_{C_1}(t) - V_{C_2}(t) \Rightarrow v(t) = v(0)e^{-t/RC_{eq.}} = V_0 e^{-2t/RC}$$

$$V_{C_1}(t) = \frac{V_0}{2} + \frac{v(t)}{2} \Rightarrow \frac{V_0}{2}\left(1 + e^{-2t/RC}\right)$$

$$V_{C_2}(t) = \frac{V_0}{2} - \frac{v(t)}{2} \quad \frac{V_0}{2}\left(1 - e^{-2t/RC}\right)$$